Lei Fan

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Summary

Applied Scientist at Amazon Alexa AI developing multimodal perception for personalized assistants, with work spanning screen understanding, egocentric video, visual memory, and model post-training. Ph.D. from Northwestern University with research in embodied perception, open-vocabulary recognition, uncertainty estimation, and 3D scene understanding.

Education

Northwestern University, Ph.D. 2019 – 2024
Computer Vision Lab; Advisor: Prof. Ying Wu

Sun Yat-sen University, B.S. & M.S. in Computer Science 2013 – 2017; 2017 – 2019

Research Expertise

- **Multimodal AI:** Video question answering, visual memory, vision-language instruction fine-tuning, and training and evaluation data development.
- **Embodied and 3D perception:** Active viewpoint selection, visual localization, depth and surface-normal estimation, and scene reconstruction.
- **Robust recognition:** Open-vocabulary and open-set recognition, evidential learning, and uncertainty quantification to distinguish confusion among known classes from ignorance of unknown inputs.

Experience

Applied Scientist, Amazon Alexa AI – Bellevue, WA May 2024 – Present
- Co-developed a visual-understanding platform evolving from discrete question answering to long-horizon visual memory, enabling reasoning over extended video histories; capabilities are being deployed in production.
- Built perception capabilities for tablet on-screen contexts and egocentric video from smart wearable devices.
- Develop multimodal perception for Alexa+ On-Screen Intelligence and live sessions, enabling personalized assistance grounded in screen content and changing visual context.
- Post-trained multimodal assistant models through instruction fine-tuning on general-purpose and domain-specific vision-language data.
- Created synthetic and human-annotated vision-language datasets for instruction fine-tuning and evaluation of multimodal perception and question answering.

Applied Scientist Intern, Amazon Robotics – Seattle, WA Jun 2023 – Sep 2023
- Developed a monocular scene-reconstruction pipeline for robotic placement, recovering scene geometry and identifying stable support surfaces from a single image.
- Built a multi-task perception model for depth estimation, surface-normal prediction, and scene segmentation.

Research Intern, Wormpex AI Research – Bellevue, WA Jun 2022 – Sep 2022
- Proposed an evidential method to distinguish confusion among known classes from ignorance of unknown inputs, enabling flexible predictions and open-set rejection.
- First-authored the resulting paper on flexible visual recognition, published at ICCV 2023.

Research Intern, Yosion Analytics – Chicago, IL Jun 2020 – Sep 2020
- Developed camera-based pallet recognition and localization for autonomous forklift guidance, and real-time empty pallet-space detection under varying lighting conditions.

Selected Publications

[1] *Beyond the Timeline: Augmenting Long-Video Memory with Grounded Entity Biographies*. Hui Ren, **Lei Fan**, Henry Pao, Han Guo, Zeeshan Zia, Ying Chen, Alexander Schwing, Gang Hua. **arXiv, 2026**.

[2] *Dictionary-Aligned Concept Control for Safeguarding Multimodal LLMs*. Jinqi Luo, Jinyu Yang, Tal Neiman, **Lei Fan**, Bing Yin, Son Tran, Mubarak Shah, René Vidal. **CVPR 2026**.

[3] *GPVK-VL: Geometry-Preserving Virtual Keyframes for Visual Localization under Large Viewpoint Changes*. Yunxuan Li, **Lei Fan**, Xiaoying Xing, Jianxiong Zhou, Ying Wu. **CVPR 2025**.

[4] *Learning to Ask Denotative and Connotative Questions for Knowledge-based VQA*. Xiaoying Xing, Peixi Xiong, **Lei Fan**, Yunxuan Li, Ying Wu. **Findings of EMNLP 2024**.

[5] *Active Open-Vocabulary Recognition: Let Intelligent Moving Mitigate CLIP Limitations*. **Lei Fan**, Jianxiong Zhou, Xiaoying Xing, Ying Wu. **CVPR 2024**.

[6] *Evidential Active Recognition: Intelligent and Prudent Open-World Embodied Perception*. **Lei Fan**, Mingfu Liang, Yunxuan Li, Gang Hua, Ying Wu. **CVPR 2024**.

[7] *Flexible Visual Recognition by Evidential Modeling of Confusion and Ignorance*. **Lei Fan**, Bo Liu, Haoxiang Li, Ying Wu, Gang Hua. **ICCV 2023**.

[8] *Unsupervised Depth Completion and Denoising for RGB-D Sensors*. **Lei Fan**, Yunxuan Li, Chen Jiang, Ying Wu. **ICRA 2022**.

[9] *FLAR: A Unified Prototype Framework for Few-sample Lifelong Active Recognition*. **Lei Fan**, Peixi Xiong, Wei Wei, Ying Wu. **ICCV 2021**.

Selected Research Projects

Grounded Entity Biographies for Long-Video Memory *arXiv 2026*

- Co-authored GEB, a memory framework that links observations of the same physical entity across video clips and retrieves entity histories alongside episodic evidence for question answering.
- The framework achieves **72.0% accuracy on EgoLifeQA**, exceeding the prior published best by 4.4 percentage points, with evaluation across four long-video QA benchmarks.

Dictionary-Aligned Concept Control for Multimodal Safety *CVPR 2026*

- Co-authored DACO, which aligns sparse-autoencoder features with a dictionary of roughly 15,000 visual concepts to steer frozen multimodal LLMs at inference time.
- Evaluated across QwenVL, LLaVA, and InternVL, the method improves safety on MM-SafetyBench and JailBreakV while preserving general model capabilities.

Geometry-Preserving Visual Localization *CVPR 2025*

- Co-developed GPVK-VL for 6-DoF camera localization under large viewpoint changes, using 2D Gaussian Splatting with confidence-aware depth and normal priors to generate virtual keyframes.
- Integrated virtual views with NetVLAD retrieval, SuperGlue matching, and PnP-RANSAC pose estimation to improve localization robustness over HLoc, InLoc, and virtual-keyframe baselines.

Active Open-Vocabulary Recognition *CVPR 2024*

- Proposed active viewpoint selection and semantic-agnostic feature fusion to address CLIP's sensitivity to occlusion and viewpoint changes in embodied environments.
- Improved ShapeNet open-vocabulary recognition accuracy from **29.6% to 53.3%** without fine-tuning CLIP; additionally evaluated the approach in Habitat.

Academic Service and Honors

Area Chair: CVPR 2027 (upcoming).
Reviewer: CVPR, ICCV, ECCV, NeurIPS, WACV, ICRA, IROS, IV, ICPR, T-PAMI, IJCV.
Honors: Northwestern Murphy Fellowship (2019); IV Best Student Paper (2018).